

```

public class MaxElementFinder {
    public static int findMax(int[] arr) {
        // Handle edge case for empty array
        if (arr == null || arr.length == 0) {
            throw new IllegalArgumentException("Array must not be empty");
        }

        // Initialize max with the first element
        int max = arr[0];

        // Traverse the array from the second element
        for (int i = 1; i < arr.length; i++) {
            if (arr[i] > max) {
                max = arr[i]; // Update max if a larger element is found
            }
        }
        return max;
    }

    public static void main(String[] args) {
        int[] numbers = {12, 34, 5, 78, 2, 45};
        System.out.println("Maximum element: " + findMax(numbers)); // Outputs 78
    }
}

```